

ALL CYCLE LEVELS

SERM PROTOCOLS

HCG GUIDE

BLOOD WORK

POST CYCLE THERAPY COMPLETE BIBLE

The Definitive Guide to HPTA Recovery, SERMs, HCG & Natural T Restoration

Nolvadex • Clomid • Enclomiphene • HCG • Natural Recovery

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DISCLAIMER: Educational purposes only. This guide discusses pharmacological protocols for harm-reduction. Consult a licensed endocrinologist before using any SERM, hormone, or related compound.

■ Understanding the HPTA — How Testosterone is Regulated

The Hypothalamic-Pituitary-Testicular Axis (HPTA) is the hormonal feedback system that controls natural testosterone production. Anabolic steroids suppress this axis — PCT is the strategy to restart it.

HPTA Component	Location	Function	What AAS Does to It
Hypothalamus	Brain	Releases GnRH (Gonadotropin-releasing hormone) in pulses	Exogenous androgens suppress GnRH release via negative feedback
Anterior Pituitary	Brain	Releases LH + FSH in response to GnRH	LH and FSH drop to near zero on cycle
Leydig Cells	Testes	Produce testosterone in response to LH	Atrophy without LH stimulation — reduced testicular size
Sertoli Cells	Testes	Produce sperm in response to FSH	Spermatogenesis suppressed — temporary infertility
Inhibin B	Testes	Negative feedback on FSH	Levels drop on cycle — FSH suppression compounded
Sex Hormone Binding Globulin (SHBG)	Liver	Binds testosterone — regulates free T	AAS lower SHBG — higher free T ratio on cycle, drops post-cycle

■ Recovery timeline is NOT linear. LH/FSH often recover within 4–6 weeks but total testosterone can take 3–6 months to fully normalise. Sperm count can take 6–18 months to recover. Pre-cycle fertility testing is strongly advised if having children is a consideration.

■ SERM Comparison — Nolvadex vs Clomid vs Enclomiphene

SERM	Mechanism	Half-Life	LH Stimulation	FSH Stimulation	Side Effects	Dose
Nolvadex (Tamoxifen)	Competitive E2 antagonist at pituitary and hypothalamus	5–7 days	Moderate	Moderate	Hot flushes, vision issues (rare), nausea	20–40 mg/day
Clomid (Clomiphene)	Mixed E2 agonist/antagonist — acts primarily at hypothalamus	5–7 days	Strong	Strong	Vision disturbances, emotional blunting, depression	25–50 mg/day
Enclomiphene	Pure E2 antagonist isomer of Clomid — no agonist activity	Short	Strong	Strong	Minimal — cleaner than Clomid	12.5–25 mg/day
Toremifene	Similar to Nolvadex — also blocks E2 in breast tissue	5 days	Moderate	Moderate	Similar to Nolvadex	60 mg/day
Raloxifene	Selective — strong bone/breast E2 blocker	28 hours	Moderate	Low	Hot flushes, rare DVT risk	60 mg/day

■ SERM Dosing Protocols by Cycle Severity

Cycle Type	SERM Choice	Week 1–2	Week 3–4	Week 5–6	Week 7–8 (if needed)
Light (Test only, short)	Nolvadex only	40 mg/day	20 mg/day	10 mg/day	—
Standard (Test only, 10–12wk)	Nolvadex ± Clomid	Nolva 40 + Clomid 50	Nolva 40 + Clomid 50	Nolva 20 + Clomid 25	Nolva 20
Heavy (Test + 2nd compound)	Nolvadex + Clomid	Nolva 40 + Clomid 50	Nolva 40 + Clomid 50	Nolva 20 + Clomid 25	Nolva 20 + Clomid 25
Very heavy (Tren/Deca based)	Nolva + Clomid + HCG pre-PCT	HCG first 10 days, then SERMs	Nolva 40 + Clomid 50	Nolva 40 + Clomid 50	Nolva 20 + Clomid 25
B&C; to natural	Full protocol	HCG 1000IU EOD × 3 weeks	Nolva 40 + Clomid 50	Nolva 40 + Clomid 25	Nolva 20

■ HCG (Human Chorionic Gonadotropin) — The Complete Guide

HCG mimics Luteinising Hormone (LH) and directly stimulates Leydig cells to produce testosterone. It prevents testicular atrophy during a cycle and can dramatically accelerate recovery if used correctly.

HCG Protocol	Timing	Dose	Frequency	Purpose
On-cycle maintenance	Throughout entire cycle	250–500 IU	2–3x per week	Prevents testicular atrophy — maintains intratesticular testosterone
Pre-PCT blast (Harpenden protocol)	Final 3 weeks before PCT	1000 IU	EOD (every other day)	Restimulates Leydig cells before SERM therapy
Pre-PCT high dose	Final 10 days before PCT	1500–2500 IU	EOD	Maximum Leydig cell stimulation — deep suppression cases
During PCT (NOT recommended)	Concurrent with SERMs	—	—	HCG suppresses the HPTA at pituitary while SERMs stimulate it — counterproductive
Fertility protocol	During or post cycle	500–1000 IU	3x/week	Maintains sperm production — essential if trying to conceive

■■ **IMPORTANT:** HCG raises intratesticular testosterone and oestradiol. Using it during PCT while also using SERMs creates contradictory signals. The gold standard is: HCG blast → wait 3 days → start SERMs.

■ Natural Testosterone Recovery Timeline

Recovery Phase	Timeframe	LH/FSH Status	Testosterone Level	What to Expect
HPTA restart initiation	Weeks 1–4 PCT	LH/FSH rising from near-zero	10–25% of pre-cycle baseline	Fatigue, low mood, libido loss — this is normal
Early recovery	Weeks 4–8	LH/FSH reaching 50–70% normal	25–60% of baseline	Energy returning, morning erections resuming
Mid recovery	Weeks 8–16	LH/FSH near normal range	60–85% of baseline	Mood stabilising, gym performance partially back
Full recovery	Weeks 12–24+	LH/FSH normal	85–100% of baseline	Depends on cycle length, compounds, and individual genetics
Extended recovery (severe cases)	6–18 months	Slow but progressive	May take 12+ months	Long cycles, deep suppression, multiple years of blasting

■ Post-Cycle Blood Work — The Full Testing Protocol

Test	Pre-Cycle Baseline	End of Cycle	Post-PCT (4 wks after)	Recovery Target
Total Testosterone	Record personal baseline	Will be suppressed	Should be rising	Back to or above personal baseline
Free Testosterone	Record baseline	Suppressed	Rising	Within normal physiological range
LH	Record baseline	Near zero	Rising — should be >2 IU/L	Back to baseline (2–12 IU/L)
FSH	Record baseline	Near zero	Rising	Back to baseline (1–12 IU/L)
Oestradiol (E2 sensitive)	Record baseline	Elevated on cycle	Normalising	Should mirror T recovery
SHBG	Record baseline	Reduced on cycle	Rising	Back toward baseline
Prolactin	If Deca/Tren used	Elevated	Should normalise	<15 ng/mL
PSA	If 35+ — record baseline	May be elevated	Check if elevated persists	Return to baseline
Full blood count	Record haematocrit	Elevated — RBC high	Normalising	Haematocrit <48–50%
Lipid panel (HDL/LDL)	Record HDL baseline	HDL crashed	Recovering	HDL should recover within 3–6 months off cycle
Sperm count (if fertility concern)	Record pre-cycle	Severely suppressed	Partial recovery	Full recovery: 3–18 months

■ Signs of Successful PCT Recovery

Indicator	Good Sign	Concerning Sign	Action if Concerning
Libido	Returns within 4–8 weeks	Still absent after 10+ weeks	Check LH/FSH/T + prolactin
Morning erections	Resume within 4–6 weeks	Absent after 8+ weeks	Assess E2, T, prolactin
Mood	Gradually stabilises	Persistent depression (12+ weeks)	Rule out low T — possible TRT consult
Energy levels	Return to near normal within 6 weeks	Chronically fatigued	Check thyroid, cortisol, iron levels too
Gym performance	80–90% of cycle peak within 6–8 weeks	Still significantly weaker	Normal if longer cycle — patience required
Blood work	LH/FSH rising, T rising	Flat LH/FSH after 8 weeks PCT	Consider extended SERM protocol or TRT evaluation

■ Natural Recovery Support Stack — Full Protocol

Supplement	Dose	Evidence Level	Primary Role	Take With
Ashwagandha (KSM-66)	600 mg/day	Strong (RCTs)	Reduces cortisol, raises LH/T by 15–20%	Food
Tongkat Ali (LJ100)	200 mg/day	Moderate	LH stimulation, free testosterone via SHBG	Morning
Zinc (picolinate/bisglycinate)	30–50 mg/day	Strong	Testosterone synthesis cofactor — aromatase inhibitor	Evening (not with calcium)
Vitamin D3	5000–10000 IU/day	Strong	Testosterone synthesis, LH signalling	Fat-containing meal, with K2
Vitamin K2 (MK-7)	100–200 mcg/day	Moderate	Calcium metabolism, testosterone support	With D3
Magnesium Glycinate	400–600 mg/day	Strong	Reduces SHBG, improves sleep, testosterone cofactor	Before bed
Boron	10 mg/day	Moderate	Reduces SHBG, raises free testosterone acutely	With meal
D-Aspartic Acid	3g/day (4 weeks)	Moderate (short-term)	LH stimulation — short-term T boost	Morning fasted
Fadogia Agrestis	600 mg/day	Early (promising)	LH mimetic — increases testosterone in studies	Morning
Omega-3 (EPA/DHA)	4g/day	Strong	Anti-inflammatory, reduces SHBG, HDL recovery	With meals
B-vitamins complex	2x RDA	Strong	Energy metabolism, hormone synthesis support	Morning with food
Pregnenolone	25–50 mg/day	Limited	Precursor to testosterone and DHEA production	Morning